

CHARCUBE · DESIGN DECISION RECORD

CharCube TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CCB-DDR-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/charcube		
Source	docs/decisions/0001-trl2-review-decisions.md @ 6ac9647		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); items 11 to 16 decided, item 9 stays open
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 8 and 10; items 11 to 16 accepted on 2026-09-25 and recorded in CCB-DDR-002); item 9 remains proposed, awaiting Amish

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed ten items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." The same instruction approved cross-cutting SwapCell interface items, a pricing rule for shared SwapCell packs, and a rule that community designs pick co-design partners per area later.

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 section) and CCB-PRC-001 v0.2. They are not repeated here.

■ Decision

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Heat recovery type	Decided by Amish, 2026-09-25: go with recommendation. Open-vented annular water jacket around the flue (option a), not the copper coil of the scaffold	CCB-PRC-001 v0.3, <code>cad/src/model.py</code> item 8
2	Budget	Decided by Amish, 2026-09-25: go with recommendation. Keep <code>budget_usd: 300</code> for the kiln parts (option a). Safety equipment (about \$40) is listed separately, is required, and sits outside the kiln budget	<code>project.yaml</code> (unchanged), CCB-REQ-001 R10, <code>bom/bom-notes.md</code>
3	Kiln type	Decided by Amish, 2026-09-25: go with recommendation. Nested-drum retort, not a Kon-Tiki flame-curtain kiln	CCB-PRC-001 v0.3
4	Target feedstock	Decided by Amish, 2026-09-25: go with recommendation. Bundled straw and mixed stalks and cobs first; loose straw only with a press, which is a separate project	CCB-REQ-001 R1 (redefined), CCB-PRB-001 v0.3
5	Burner throat with preheated secondary air	Decided by Amish, 2026-09-25: go with recommendation. Include it. Hole sizing at TRL 3 is in CCB-CAL-001 section 8 and leads to open item 11	CCB-PRC-001 v0.3, CCB-CAL-001
6	Temperature logger as standard	Decided by Amish, 2026-09-25: go with recommendation. Include it (about \$32) as the batch record	<code>bom/bom.csv</code> line 13, CCB-REQ-001 R9
7	Insulation	Decided by Amish, 2026-09-25: go with recommendation. 25 mm ceramic fibre blanket for the prototype	<code>bom/bom.csv</code> line 11
8	Jacket support	Decided by Amish, 2026-09-25: go with recommendation. Tripod lifted aside by two people for loading	<code>cad/src/model.py</code> item 10, CCB-REQ-001 R11
10	Name	Decided by Amish, 2026-09-25: go with recommendation. Keep the name CharCube; no reshaping of the concept	<code>project.yaml</code> (unchanged)

The pitch and problem lines had no recommended rewording in the TRL 2 review and are unchanged.

Cross-cutting approvals from the same instruction, recorded here:

- **SwapCell interface v0.3.** Decided by Amish, 2026-09-25 (cross-cutting): the SwapCell interface adds a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating for vehicles. CharCube does not use SwapCell (the logger runs from a USB power bank), so no v0.3 items apply.
- **Shared packs are priced once.** Decided by Amish, 2026-09-25 (cross-cutting). Not applicable: CharCube has no SwapCell pack.
- **Co-design partners per area later.** Decided by Amish, 2026-09-25 (cross-cutting): community designs pick co-design partners per area later. Partners stay open.

Items that remain open

Table 2. Items open at v0.1. Items 11 to 16 were decided by Amish on 2026-09-25 (see CCB-DDR-002); item 9 remains open.

#	Item	Status and recommendation
9	First region, residue and partner for co-design	Proposed, awaiting Amish. No recommendation was made (the review gave examples only: paddy straw in northwest India, or maize and cotton stalks in East Africa). Portfolio rule: partners are chosen per area later
11	Secondary air manifold form	Decided by Amish, 2026-09-25: go with recommendation (CCB-DDR-002). CCB-CAL-001 shows the TRL 2 25 mm pipe ring would need about 46 Pa to pass the peak secondary air against about 8 Pa of available suction. The TRL 3 model uses a 230 mm sleeve (shroud) around the throat with an open bottom and band damper, and 24 holes of 12 mm. Recommendation: accept the shroud
12	R6 hot water shortfall (6.4 MJ against 10 MJ)	Decided by Amish, 2026-09-25: go with recommendation (CCB-DDR-002). Options: (a) add a spiral baffle insert in the jacket sleeve and 25 mm of blanket on the jacket shell (about 13.9 MJ on paper, about \$15 more); (b) relax R6 to 5 MJ; (c) drop the jacket, as TRL 2 option (c). Recommendation: (a), checked at TRL 4 if Amish lifts the hold, since the insert also adds tar and soot fouling to clean
13	R8 start-up wood (5.8 kg central against 5 kg)	Decided by Amish, 2026-09-25: go with recommendation (CCB-DDR-002). Options: blanket the lid and top band (saves about 1.4 kW of loss), run at lower excess air, or relax R8 to 8 kg. Recommendation: blanket the lid (about \$6) and keep the 5 kg target until batches are logged
14	R5 burn time (4.5 h central against 4 h)	Decided by Amish, 2026-09-25: go with recommendation (CCB-DDR-002). Heat conduction into the packed charge sets the burn time. Options: relax R5 to 5 h, or reduce the conduction path with a perforated central tube in the retort. Recommendation: relax R5 to 5 h light to end of flaming; the 16 h unload limit is met
15	R9 accuracy above 700 °C	Decided by Amish, 2026-09-25: go with recommendation (CCB-DDR-002). The MAX31855 accuracy is specified to 700 °C. Recommendation: keep class 1 probes and restate R9 as ± 5 °C to 700 °C and indicative above
16	Kiln operating notes versus build notes	Decided by Amish, 2026-09-25: go with recommendation (CCB-DDR-002). The TRL 2 review said the safety kit would be "stated as required in the build notes at TRL 3". Build notes are TRL 4 material, so the requirement is stated in CCB-REQ-001 R10 and <code>bom/bom-notes.md</code> instead. Recommendation: accept

■ Consequences

- CCB-PRB-001, CCB-PRC-001 and CCB-REQ-001 move to v0.3 with these decisions; the precis no longer lists items 1 to 8 and 10 as proposed.
- CCB-REQ-001 R1 applies to the decided feedstock (packed or bundled), and R10 now states that the \$300 budget covers kiln parts, with the safety kit required and listed separately.
- The TRL 3 model, drawing CCB-DWG-001 and CCB-CAL-001 follow the decided retort, jacket, throat, logger, blanket and tripod.
- TRL 4 work (building, testing, purchasing) is on hold by Amish's instruction, whatever the outcome of the open items.